RouteX: Role-Based Access Control (RBAC) & Use Case Diagram Reference Guide

Project: RouteX — Logistics Delivery Operations & Truck Profitability Management System
Client / Domain Organization: Redeemed Route Logistics Inc. (RRLI)
Target Audience: Systems Analyst / UI-UX Designer / Software Engineering Groupmate
Document Purpose: Complete specification of Actors, Roles, Permissions, Use Cases, and UML Relationships to construct an accurate and compliant **Use Case Diagram**.

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1. System Overview & Scope Boundaries

RouteX is a centralized logistics web application designed specifically for **Redeemed Route Logistics Inc. (RRLI)** to monitor completed delivery trips, audit financial revenue and expenses, manage cold chain (refrigerated) and ambient fleet vehicles, verify proof-of-delivery (POD) documentation, and generate operational summaries.

System Boundaries & Invariants

- **In Scope:**
  - Completed delivery records and multi-store drop-off tracking.
  - Granular financial breakdown (revenue, fuel, tolls, driver/helper rates, allowances, entry fees, miscellaneous, net profit, margin).
  - Truck fleet management (Owned 4/6/10-wheelers, Subcontracted fleets, cold storage vs. dry cargo capabilities).
  - Staged Excel spreadsheet bulk importation and validation.
  - Digital Proof of Delivery (POD) document viewing (images, PDFs) and attachment.
  - Custom report template authoring and CSV/Excel data export.
  - User identity lifecycle, Argon2id security, password recovery, and immutable audit logging.
- **Out of Scope (Explicit Domain Boundaries):**
  - **No Live GPS / Dispatching:** RouteX is an operations ledger and profitability auditor, not a real-time driver GPS tracking tool.
  - **No Hard Deletes:** Data records (deliveries, trucks, users) are preserved for auditing; user accounts use soft deactivation (`isActive: false`).
  - **No Open Self-Registration:** All system accounts are provisioned by an Administrator; guest users cannot register themselves.

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2. Actors & Role Hierarchy

A Use Case Diagram requires clearly identified **Actors** (human or external systems interacting with the system) and their **generalization / inheritance** relationships.

2.1 The Six Canonical User Roles

RouteX implements **6 Canonical Business Roles** mapped into **3 Tiered Access Levels**:

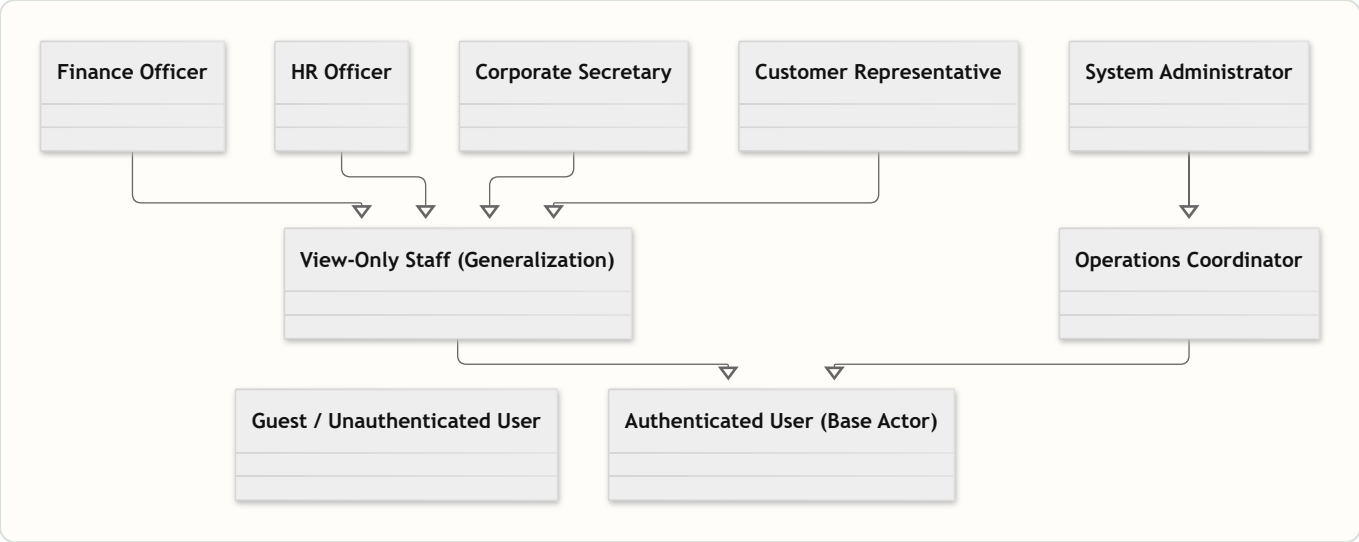
Role Name	Access Level	UI Label & Badge Tone	Primary Business Purpose & Domain Responsibilities
`admin`	**Admin** (`admin`)	`Admin` *(Primary / Green)*	Full system administration, user account management, role assignment, credential resets, audit log inspection, report template customization, and full operational write capabilities.
`operations`	**Operations** (`operations`)	`Operations` *(Primary / Green)*	Daily logistics operations: creating/editing deliveries, recording multi-store drop-offs, uploading and reviewing Excel bulk trips, attaching/replacing POD files, registering fleet trucks, and exporting reports.
`finance`	**View-Only** (`view_only`)	`Finance` *(Neutral)*	Financial auditing: inspects trip revenues, operational expenses (fuel, tolls, helper pay), profit margins, truck profitability rankings, and financial reports.
`hr`	**View-Only** (`view_only`)	`HR` *(Neutral)*	Personnel auditing: inspects delivery records, driver and helper crew assignments, daily trip wage rates, and meal allowances to prepare payroll summaries.
`secretary`	**View-Only** (`view_only`)	`Secretary` *(Neutral)*	Operational compliance: monitors delivery completion, ensures POD documents are attached and complete, checks waybill status, and reviews company reports.
`customer_representative`	**View-Only** (`view_only`)	`Customer Rep` *(Neutral)*	Customer service inquiry: inspects client delivery waybills, store drop-off timelines, delivery dates, and POD receipt verification to resolve customer claims.

2.2 Unauthenticated & Secondary Actors

- `Guest / Unauthenticated User` : Any visitor accessing the public route (`/login`). Can submit credentials to log in, complete first-login setup, or perform password self-recovery.
- `System Audit Logger` **(Secondary / Supporting Actor)**: Automated background subsystem that intercepts administrative mutations and records immutable attribute transitions (old vs. new values).

2.3 Actor Generalization Hierarchy (UML Inheritance)

In UML Use Case Diagrams, you can reduce clutter by modeling **Actor Inheritance**:



- **Inheritance Rules:**
  - `View-Only Staff` inherits all general viewing capabilities (`Dashboard`, `Deliveries`, `PODs`, `Trucks`, `Reports`, `Settings`).
  - `Finance`, `HR`, `Secretary`, and `Customer Representative` are concrete specializations of `View-Only Staff` (they share the same system permissions, but represent distinct organizational goals).
  - `Operations Coordinator` inherits viewing capabilities and adds write/mutation capabilities for operational modules.
  - `System Administrator` inherits all `Operations` capabilities and adds administrative privileges (`User Management`, `Audit Logs`, `Template Authoring`).

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3. Comprehensive Use Case Catalog by Subsystem

Below is the complete list of system use cases, their descriptions, primary actors, and UML relationships (`<<include>>` or `<<extend>>`).

Subsystem 1: Identity & Access Management (IAM)

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-01	**Log In**	Guest User	—	Authenticate into RouteX using username/email and password.
UC-02	**Complete First-Login Setup**	Authenticated User	`<<extend>>` UC-01 *(Condition: temporary password assigned)*	Mandatory workflow on first login: set a new permanent password and register security questions.
UC-03	**Recover Password via Security Question**	Guest User	`<<extend>>` UC-01 *(Condition: user forgot password)*	Self-service password recovery by answering the predefined security question.

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-04	**Manage Account Profile**	Authenticated User	`<<include>>` UC-01	User updates own display name, username, password, or uploads profile picture avatar.
UC-05	**Log Out**	Authenticated User	`<<include>>` UC-01	Terminate the active session and invalidate HMAC-SHA256 session cookie.

Subsystem 2: Delivery Operations & Trip Management

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-06	**View Delivery Ledger**	View-Only Staff, Operations, Admin	`<<include>>` UC-01	View completed delivery records in a tabular ledger with pagination and financial totals.
UC-07	**Search & Filter Deliveries**	View-Only Staff, Operations, Admin	`<<include>>` UC-06	Filter deliveries by global date range, client, truck plate, driver, or POD attachment status; sort via 3-state sort controls.
UC-08	**View Delivery Details Drawer**	View-Only Staff, Operations, Admin	`<<include>>` UC-06	Open the detailed slide-out drawer to view trip overview, crew, vehicle specs, drop-offs, and financial breakdown.
UC-09	**Create Delivery Record**	Operations, Admin	`<<include>>` UC-01, `<<include>>` UC-10, `<<include>>` UC-11	Register a new delivery trip manually.
UC-10	**Record Multi-Store Drop-off Sequence**	Operations, Admin	`<<include>>` UC-09, `<<include>>` UC-12	Specify ordered destination stores, drop-off locations, and individual delivery quantities.
UC-11	**Assign Vehicle & Crew**	Operations, Admin	`<<include>>` UC-09, `<<include>>` UC-12	Select truck plate (owned/subcontracted), assign driver, and assign up to two helpers with respective wage rates.
UC-12	**Edit Delivery Record**	Operations, Admin	`<<include>>` UC-08	Modify operational or financial attributes of an existing delivery.
UC-13	**Calculate Trip Financials & Margin**	System / Automated	`<<include>>` UC-09, `<<include>>` UC-12	Automatically compute revenue, total expenses, net income, and profit margin percent.

Subsystem 3: Batch Staging & Excel Bulk Import

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-14	**Upload Excel Trip File**	Operations, Admin	`<<include>>` UC-01	Upload `.xlsx` / `.csv` spreadsheet containing multiple delivery trip rows into the staging environment.
UC-15	**Review & Validate Staged Deliveries**	Operations, Admin	`<<include>>` UC-14	Inspect parsed rows, check for missing fields, rate anomalies, or unrecognized truck plates prior to saving.
UC-16	**Confirm & Commit Batch Import**	Operations, Admin	`<<include>>` UC-15	Atomically commit validated staging rows into the active delivery database.

Subsystem 4: Proof of Delivery (POD) & Document Management

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-17	**View Proof of Delivery (POD)**	View-Only Staff, Operations, Admin	`<<include>>` UC-08	View attached POD files (high-resolution PNG/JPG receipts or embedded PDF documents).
UC-18	**Download POD File**	View-Only Staff, Operations, Admin	`<<include>>` UC-17	Download the original signed delivery receipt or invoice document.
UC-19	**Upload / Replace POD Document**	Operations, Admin	`<<extend>>` UC-17 *(Condition: missing or updated document)*	Attach or update a digital POD file against a delivery stop.

Subsystem 5: Fleet & Truck Profitability Analytics

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-20	**View Fleet & Profitability Grid**	View-Only Staff, Operations, Admin	`<<include>>` UC-01	Review owned truck cards and subcontracted fleet vehicles with aggregated net profits and performance KPIs.
UC-21	**Filter & Rank Trucks**	View-Only Staff, Operations, Admin	`<<include>>` UC-20	Filter by cargo storage capability (Refrigerated/Cold Chain vs. Dry Cargo), wheel count, or performance tier; rank by profit, revenue, or margin.
UC-22	**View Truck Operational Profile**	View-Only Staff, Operations, Admin	`<<include>>` UC-20	View dedicated truck profile page containing specifications, vehicle photo, revenue history, and assigned trip log.
UC-23	**Register / Update Truck Information**	Operations, Admin	`<<include>>` UC-01	Register a new truck plate, define classification, ownership (Owned/Subcon), cargo capability, and upload vehicle image.

Subsystem 6: Operational & Financial Reporting

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-24	**Generate Operational Reports**	View-Only Staff, Operations, Admin	`<<include>>` UC-01	View structured reports across built-in layouts: Client Summary, Truck Summary, or Trip Ledger.
UC-25	**Filter Report by Global Date Range**	View-Only Staff, Operations, Admin	`<<include>>` UC-24	Recompute report totals dynamically using date presets (Today, This Week, This Month, All Time) or custom dates.
UC-26	**Export Report to CSV / Excel**	Operations, Admin	`<<include>>` UC-24	Export RFC 4180-compliant CSV containing metadata, active filters, column headings, and calculated summary totals.
UC-27	**Print / Save Report as PDF**	View-Only Staff, Operations, Admin	`<<include>>` UC-24	Open clean browser print dialogue optimized for formal paper or PDF archiving.
UC-28	**Author Custom Report Template**	Admin	`<<extend>>` UC-24 *(Condition: Admin customizes layout)*	Customize column visibility, display labels, ordering, and aggregation groupings.
UC-29	**Delete Custom Report Template**	Admin	`<<include>>` UC-28	Delete user-created templates (built-in system default templates cannot be deleted).

Subsystem 7: User Administration & System Auditing

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-30	**View User Directory**	Admin	`<<include>>` UC-01	View list of system staff, assigned roles, email addresses, and account status (`Active` vs. `Inactive`).
UC-31	**Provision New User Account**	Admin	`<<include>>` UC-30	Create account with temporary credentials and assign canonical role.
UC-32	**Modify User Role & Status**	Admin	`<<include>>` UC-30	Change staff role assignment or toggle active status (soft-disable without deleting records).
UC-33	**Reset User Password (Admin-Assisted)**	Admin	`<<include>>` UC-30	Issue temporary password to unlock an account when user forgets security question answers.
UC-34	**Inspect System Audit Logs**	Admin	`<<include>>` UC-30	View audit trail capturing security events, credential updates, and record diffs showing old vs. new state changes.

4. RBAC Permissions Matrix (CRUD & Execution Mapping)

This matrix maps system permissions directly to the **6 Canonical Roles**:

Module / Permission Key	Code Identifier	Admin	Operations	Finance	HR	Secretary	Customer Rep
Dashboard							
View Dashboard & KPIs	`dashboard:view`	✅	✅	👁 (Read)	👁 (Read)	👁 (Read)	👁 (Read)
Deliveries							
View Delivery Ledger & Details	`deliveries:view`	✅	✅	👁 (Read)	👁 (Read)	👁 (Read)	👁 (Read)
Create New Delivery	`deliveries:create`	✅	✅	❌	❌	❌	❌
Edit Existing Delivery	`deliveries:edit`	✅	✅	❌	❌	❌	❌
Complete / Finalize Trip	`deliveries:complete`	✅	✅	❌	❌	❌	❌
Manage Store Drop-offs	`dropoffs:manage`	✅	✅	❌	❌	❌	❌
Excel Batch Staging							
Upload Excel File	`imports:upload`	✅	✅	❌	❌	❌	❌
Review & Confirm Import	`imports:review_confirm`	✅	✅	❌	❌	❌	❌
Proof of Delivery (POD)							
View & Download POD	`pod:view`	✅	✅	👁 (Read)	👁 (Read)	👁 (Read)	👁 (Read)
Upload / Replace POD	`deliveries:edit`	✅	✅	❌	❌	❌	❌
Fleet & Profitability							
View Trucks & Profitability	`profitability:view`	✅	✅	👁 (Read)	👁 (Read)	👁 (Read)	👁 (Read)
Register / Edit Trucks & Photos	`profitability:manage`	✅	✅	❌	❌	❌	❌
Reports							
View Report Document Previews	`reports:view`	✅	✅	👁 (Read)	👁 (Read)	👁 (Read)	👁 (Read)

Module / Permission Key	Code Identifier	Admin	Operations	Finance	HR	Secretary	Customer Rep
Export Reports to CSV/Excel	`reports:export`	✅	✅	❌	❌	❌	❌
Author / Delete Custom Templates	`reports:manage_templates`	✅	❌	❌	❌	❌	❌
Administration & IAM							
View User Accounts Directory	`users:view`	✅	❌	❌	❌	❌	❌
Manage Accounts, Roles & Resets	`users:manage`	✅	❌	❌	❌	❌	❌
View Audit Logs & Transitions	`users:manage`	✅	❌	❌	❌	❌	❌
View Settings Page	`settings:view`	✅	✅	👁 (Read)	👁 (Read)	👁 (Read)	👁 (Read)
Manage System Configuration	`settings:manage`	✅	❌	❌	❌	❌	❌

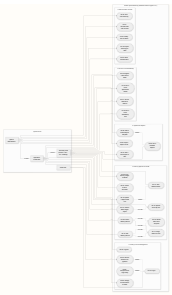
Legend:

- ✅ **Full Access:** Can view, execute, create, edit, or submit changes.
- 👁 **Read Only:** Can view data, filter, search, and inspect drawers; mutation controls are hidden and server-blocked (HTTP 403).
- ❌ **Prohibited:** Action and navigation route are completely inaccessible.

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5. Ready-to-Use Mermaid UML Diagram

Your groupmate can paste this code directly into markdown viewers, GitHub, or Mermaid Live Editor (`mermaid.live`):



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6. Role-Specific Business Context for Use Case Narratives

If your groupmate needs to write descriptions or narrative specifications for each role, use these verified domain summaries:

1. Finance Officer (`finance`)

- **Why they use RouteX:** To audit delivery profitability without risking operational tampering.
- **Key Use Cases:** UC-06 (View Deliveries), UC-08 (Delivery Details Drawer), UC-20 (Truck Profitability Grid), UC-24 (Client & Truck Summary Reports).
- **Specific Data Watched:** Fuel expenses, highway toll receipts, helper daily allowances, client delivery rates, gross revenue, net profit margin percentages, and incomplete expense flags.

2. HR Officer (`hr`)

- **Why they use RouteX:** To verify staff assignments and calculate bi-monthly crew compensation.
- **Key Use Cases:** UC-06 (Delivery Ledger), UC-08 (View Crew Details), UC-24 (Trip Ledger Report).
- **Specific Data Watched:** Driver assignments, Helper 1 and Helper 2 assignments, daily trip rates (e.g. ₱700 driver daily rate), meal allowances (e.g. ₱300), and delivery dates.

3. Corporate Secretary (`secretary`)

- **Why they use RouteX:** To verify organizational compliance, client contract fulfillment, and document completeness.
- **Key Use Cases:** UC-06 (Deliveries), UC-17 (View POD Documents), UC-20 (Fleet Registry), UC-24 (Operational Reports).
- **Specific Data Watched:** Stamped waybill receipts, customer signature attachments on PODs, vehicle registration plate types, and delivery dates.

4. Customer Representative (`customer_representative`)

- **Why they use RouteX:** To answer client delivery inquiries and resolve missing delivery disputes.
- **Key Use Cases:** UC-06 (Search Deliveries by Client/Waybill), UC-08 (Drop-off Sequence), UC-17 (View Signed POD).
- **Specific Data Watched:** Client name (e.g., Alfamart, SM Supermarket), waybill numbers, store drop-off timestamps, quantities delivered, and signed delivery receipts.

5. Operations Coordinator (`operations`)

- **Why they use RouteX:** Hands-on operational management of daily dispatch execution and data entry.
- **Key Use Cases:** UC-09 (Create Delivery), UC-12 (Edit Delivery), UC-10 (Drop-off stops), UC-14 (Excel Bulk Upload), UC-16 (Confirm Staged Batch), UC-19 (Upload PODs), UC-23 (Register Fleet Trucks), UC-26 (Export CSV).

6. System Administrator (`admin`)

- **Why they use RouteX:** System security, account provisioning, role assignment, format configuration, and accountability auditing.
- **Key Use Cases:** UC-30 through UC-34 (User Directory, Account Provisioning, Role Management, Credential Resets, Audit Logs with State Diffs), UC-28 (Custom Report Template Authoring).

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7. Diagramming Best Practices & Tips for Your Groupmate

1. **Use Actor Inheritance to Keep the Diagram Clean:**
   Instead of drawing lines from all four view-only roles (`Finance`, `HR`, `Secretary`, `Customer Rep`) to every single view use case, create a generalization actor called `View-Only Staff`. Draw arrows from `Finance`, `HR`, etc. inheriting from `View-Only Staff`, and draw single association lines from `View-Only Staff` to the viewing use cases.
2. **Proper Use of `<<include>>`:**
   Use `<<include>>` when a use case cannot complete without another. Example: *Creating a Delivery* always includes *Recording Drop-offs* and *Assigning Crew*. The arrow points **from** the base use case **to** the included use case: `(Create Delivery) .-> <<include>> (Assign Crew)`.
3. **Proper Use of `<<extend>>`:**
   Use `<<extend>>` for optional or conditional workflows. Example: *First-Login Setup* only happens conditionally if the account is new; *Upload POD* is an extension when viewing a delivery stop that lacks a document. The arrow points **from** the extending use case **back to** the base use case: `(Complete First-Login Setup) .-> <<extend>> (Log In)`.
4. **Tool Compatibility:**
   This document's structure and naming conventions can be directly copied into popular modeling tools:
   - **draw.io / diagrams.net:** Standard UML Use Case stencil set.
   - **Lucidchart:** UML Use Case library.
   - **Visual Paradigm / StarUML:** Use Case Diagram view.
   - **PlantUML:** Convert the catalog above using standard `actor` and `usecase` syntax.